Dev Mehta

AI/ML & Agentic Systems • Backend & Distributed Software • Forward-Deployed / Full-Stack • B.Tech CSE, 2027

Chennai, India • Open to relocation | +91-9080111855 | devmehta060505@gmail.com | github.com/TEKURU-ZENO
linkedin.com/in/dev-mehta-1861bb2a0 | port-folio-mine-ashy.vercel.app

Summary

Engineer (2027) who builds across the full range: **LLM, RAG and agentic systems**, **scalable backend and distributed software**, and **client-facing products** shipped end to end. I take open, undefined problems to deployed systems, care about correctness, evaluation and the user, and use AI tools daily while owning the judgement calls. Strong **Python, PyTorch and TypeScript**, with real production experience across three internships. **Four peer-reviewed publications**, an open-source merge upstream, CGPA 9.86/10, Rank 1; All India Rank 827, AINCAT (800,000+ candidates).

Technical Skills

Languages	Python (primary), Java, C/C++, Go, JavaScript / TypeScript, SQL, R, Bash
GenAI / Agentic	LLMs (OpenAI, Anthropic, HuggingFace), RAG, embeddings & vector search, agent orchestration, LangChain / LangGraph, MCP tool-calling, prompt engineering, fine-tuning, Pydantic structured outputs, grounding & evaluation
ML / Data	PyTorch, scikit-learn, transformers, NLP, classification / clustering / forecasting / anomaly detection; pandas, NumPy, Spark / PySpark, Databricks, feature engineering, model evaluation
Backend	FastAPI, Django, Node.js / Express; REST & gRPC APIs, microservices, event-driven (Apache Kafka), authentication (JWT, RBAC), payment & webhook integrations
Frontend	React, TypeScript, Angular, Redux Toolkit, Tailwind CSS, React Native, responsive UI
Infra / MLOps	Docker, Kubernetes, AWS, GCP, Azure, CI/CD, Prometheus, Grafana, OpenTelemetry, PostgreSQL, MongoDB, Git, Linux
Practices	Distributed systems, system design, DSA, testing & benchmarking, code review, observability, ownership, clear communication

Projects

HECATE, Distributed Backend & Autonomous-SRE Platform github.com/TEKURU-ZENO/HECATE

Python, Apache Kafka, Kubernetes, Prometheus, Grafana, Jaeger, PostgreSQL, Docker

- Built a **distributed system of seven services** over an Apache Kafka event bus, each independently deployable and horizontally scalable, ingesting high-throughput data with **anomaly detection** (Isolation Forest + baselines) and automated remediation.
- Added distributed tracing, metrics and a **seed-controlled evaluation harness** for reproducible benchmarking of detection and reliability under load.

SELENE-OS, Agentic AI Orchestration Platform github.com/TEKURU-ZENO/SELENE-OS

Python, FastAPI, LangGraph, MCP tool-calling, LLM APIs, React, TypeScript

- Designed an **agent architecture**: agents that plan and act over tools via **MCP**, inter-agent orchestration with LangGraph, RAG context, and a clean React front end wired to LLM APIs (GPT, Claude).
- Built in deterministic replay, explainable lineage and human-approval gates before consequential actions.

Recoup, LLM Decision Engine on Payments github.com/TEKURU-ZENO/Recoup

Python, FastAPI, PostgreSQL, LLM grounding validator, benchmarking, Docker

- Built a deterministic engine on a **payments stack** (Razorpay webhooks) with an LLM behind a **grounding validator** (0.0% ungrounded outputs), exact integer arithmetic, hard policy guards and an audit trail.
- **Validated rigorously**: 132 automated tests, a 4-arm / 20-seed benchmark with Common Random Numbers, and shadow-mode evaluation over 400 real events.

NEXUS GRID, Infrastructure Intelligence on Apache Spark portfolio

PySpark, Spark Structured Streaming, Kafka, Delta Lake, Neo4j, FastAPI, Go

- Built large-scale streaming data pipelines and **forecasting models** (PySpark ML) over high-volume operational data, with a Neo4j digital twin and a RAG copilot over the system's state.

BioGPT-DDI, Biomedical NLP Model [Published]

PyTorch, HuggingFace, BioBERT / BioGPT, fine-tuning, relation extraction, metrics

- **Fine-tuned transformer models** (BioBERT, BioGPT) for drug-drug interaction extraction to **86.64% F1** on DDI-2013, with clean train/validation/test design and error analysis. Published, IJIREEICE 2025.

PERSEPHONE, Precision-Oncology Decision Support github.com/TEKURU-ZENO/PERSEPHONE

Node.js, multi-agent recommendations, DAG orchestration, knowledge graph, evidence grounding

- Built an **evidence-grounded precision-oncology decision-support** app that models tumours as evolving systems: **multi-agent recommendations** over a knowledge graph with **DAG-based orchestration** and an explainable reasoning trail.

VeriX, Java Full-Stack Application (Full Life Cycle) github.com/TEKURU-ZENO/VeriX

Java, Node.js / Express, Angular, TypeScript, MongoDB, JWT, Docker

- **Java** Controllers/Services/Repository backend with JWT auth and RBAC, a TypeScript front end, and a network resilience simulator, designed, built, tested and deployed. Documented in a PRD, SRS and ADRs.

Real-Time Voice Agent, Telephony + LLM

Python, Whisper STT, LiveKit, Twilio / Telnyx, WebRTC, LLM tool-calling

- Built a **real-time voice agent** over live telephony: Whisper speech-to-text, LiveKit / WebRTC audio, with **interruption handling, turn-taking and mid-call tool calling**, a low-latency, client-facing system.

Experience

Regacore Remote

Technical and Creative Intern *Jun 2026 -- Present*

- Given an **open, undefined brief**: researched the approach, benchmarked candidate designs, and **owned the flagship product end to end**, including S3 document storage and backend auth, working directly with non-technical stakeholders.

Airawat Research Foundation (IIT Kanpur Centre of Excellence) Remote

Full Stack AI Engineer Intern *Jul 2026 -- Sep 2026*

- Built and deployed **full-stack AI features and RAG pipelines** behind Python APIs over messy real-world data, on cloud with Docker and CI/CD, contributing through **code review** and documentation.
- Owned data preparation and high-volume pipelines end to end, from ingestion and cleaning through model serving and monitoring.

Sri Jayaram Infotech Pvt Ltd Remote

AI/ML Developer Intern --- delivered to a live client *Jan 2026 -- Jun 2026*

- Trained, evaluated and **deployed ML and CV/OCR models into a live client platform**, debugged production issues through systematic root-cause analysis, and measured and improved performance on real data.
- Worked directly against client requirements and real delivery dates, maintaining API contracts and CI/CD.

Education, Publications & Recognition

SRM Institute of Science and Technology, Vadapalani Chennai, India

B.Tech, Computer Science and Engineering *CGPA 9.86 / 10* *Rank 1, Overall Proficiency* *2023 -- 2027*

- Coursework: Machine Learning, Data Structures & Algorithms, Operating Systems, Computer Networks, DBMS, Distributed Systems, OOP, Probability & Statistics, Linear Algebra.
- **Four peer-reviewed publications** (ATLAS, ORION, BioGPT-DDI, CEREP), biomedical NLP and applied ML. **Aperture 3.0** open-source merged upstream. **Top 1000 Builders** from 10,000+, OpenAI × Outskill (2026). Winner / podium in three front-end hackathons (2025). Also built a real-time voice agent (Whisper STT, LiveKit, Twilio/Telnyx telephony, WebRTC, interruption handling and mid-call tool calling).